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Technical Evaluation of an Offline Voice-Controlled Mobile Application for Caregiver Procedure Assistance in Older Adult Care

Eduardo Guzmán Olea [1], Juan Manuel Lopez-Tellez [2], Fernando Martell[2] Carlos Paredes-Orta [3],

[1] Centro de Investigación en Sistemas de Salud, Universidad No. 655, Colonia Santa María Ahuacatitlán, Cerrada Los Pinos y Caminera, C.P. 62100, en Cuernavaca, Morelos, México.
[2] Centro de Investigaciones en Óptica, Aguascalientes, 20200, México.
[3] Conacyt-Centro de Investigaciones en Óptica, Aguascalientes, 20200, México.
* Correspondence: cparedes@cio.mx (C.P.O.)

Abstract: This study presents a technical evaluation of an offline voice-controlled mobile application designed to assist caregivers of dependent older adults in accessing structured caregiving procedures. The application organizes 127 caregiving processes and associated supporting information in an Android environment developed with the Unity engine and a local SQLite database. The system includes caregiver and patient information, procedure navigation, images, glossary content, care records, calendar functions, and predefined voice-assisted navigation accessed through a native Android Java bridge. The experimental evaluation comprised 20 startup measurements on a POCO M5s smartphone running Android 13, 400 voice-command trials involving eight speakers, five Android devices, and ten predefined Spanish commands, a separate balanced end-to-end latency dataset of 150 successful measurements, 1,250 SQLite query executions, and 13 offline functional tests. The voice protocol produced 355 correct recognitions out of 400 trials (88.75%). The balanced end-to-end measurements showed a mean response latency of 3.604 ± 0.769 s from activation of speech recognition to procedure navigation. SQLite executions were valid in 1,250/1,250 cases, with mean query latencies ranging from 0.0985 to 1.8904 ms across the five query types. Runtime profiling on the POCO M5s yielded an overall mean PSS of 366.79 MB, while three 30-min battery sessions produced a mean reduction of 5.33 ± 0.58 percentage points. The results characterize the technical behavior of the evaluated application under disconnected conditions. They should not be interpreted as evidence of clinical effectiveness, caregiver usability, or general-purpose speech-recognition performance, and they are limited to the tested hardware, software configuration, command vocabulary, and acoustic conditions.

Keywords: mHealth; caregivers; older adults; offline mobile application; voice interaction; Android; SQLite; Mexico

Citation: Eduardo Guzmán Olea,Juan Manuel Lopez-Tellez, Fernando Martell and Carlos Paredes-Orta Technical Evaluation of an Offline Voice-Controlled Mobile Application for Caregiver Procedure Assistance in Older Adult Care. *Sensors* **2026**, *1*, 0. https://doi.org/

Received:
Accepted:
Published:

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1. Introduction

Population ageing is increasing the demand for long-term and day-to-day support systems that can help older adults maintain functional ability and receive appropriate care in their homes and communities. The World Health Organization has emphasized the need to organize health and social-care responses around functional ability and person-centred support as populations age [1,2]. WHO has also identified dementia as a major cause of disability and dependency and emphasized the need for coordinated support for affected people and their carers [3]. In Mexico, the 2020 population census reported 15.1 million people aged 60 years or older, representing 12% of the national population [4]. These demographic conditions increase the relevance of tools that can provide caregivers with accessible and structured information.

Mobile health (mHealth) applications have become a practical mechanism for delivering health information, reminders, monitoring functions, and support services.

Previous analyses of caregiver-oriented applications have shown that available tools cover a broad range of functions, but that comprehensive solutions remain relatively uncommon [5–7]. Systematic mapping and review studies have also identified usability, accessibility, content organization, and user-centred design as recurring challenges in mobile health applications for older adults [8–11]. These findings suggest that the value of a caregiving application depends not only on the amount of information provided, but also on how that information is structured and accessed during a real caregiving task.

Voice interaction offers an additional mechanism for reducing interaction effort, particularly when the caregiver's attention is occupied by another activity. Voice assistants can interpret spoken commands and provide synthesized responses [12], and recent reviews have examined their use and potential benefits for older adults [13]. At the same time, voice interaction in health applications raises usability, privacy, reliability, and environmental-noise considerations [12,14]. For a task-oriented application, a constrained vocabulary can be preferable to open-domain conversational interaction because the system only needs to identify commands associated with predefined actions.

The present work addresses these issues through the implementation and functional evaluation of a mobile application derived from a structured caregiving manual for dependent older adults. The project documentation describes 127 caregiving processes, supporting images and glossary information, caregiver and patient records, a care log, and a calendar. The application was developed in Unity for Android and uses a local SQLite database so that the caregiving content can remain available on the device. Voice-to-text and text-to-speech are integrated through native Android speech interfaces accessed from Unity through a Java bridge. The supplied Unity integration invokes the Android bridge for speech recognition and synthesis, while the application evaluates the recognized text against a constrained command vocabulary.

The objective of this study is to technically characterize the implemented application under disconnected conditions. The evaluation focuses on startup behavior, predefined voice-command recognition, end-to-end response latency, local database performance, offline functional execution, and runtime resource behavior. The study does not evaluate clinical effectiveness or caregiver usability.

2. Related Work

Mobile applications for caregivers have been examined from both functional and usability perspectives. Grossman et al. analyzed 44 caregiver-oriented applications and found that many focused on information and resources, communication, or caregiver-recipient interaction, while relatively few addressed several categories simultaneously [5]. Sala-González et al. subsequently reviewed mobile applications for informal caregivers and highlighted heterogeneity in functions, quality, and evidence supporting their use [6]. More recent work has examined interventions for family caregivers of older adults with dementia and reported potential benefits across several caregiver outcomes, while also emphasizing variation in intervention design and evidence quality [7]. Zou et al. specifically examined publicly available mHealth applications for dementia caregivers and identified variation in information quality and tailoring [8].

Studies focused on older adults provide complementary design evidence. Paiva et al. mapped mobile applications for elderly healthcare and identified recurring application categories and research gaps [9]. Liu et al. synthesized interface and persuasive-feature recommendations for older adults, emphasizing perceptual, motor, and cognitive considerations [10]. Cruz Zapata et al. reviewed empirical usability studies of mHealth applications and found that usability evaluation methods vary considerably across studies [11]. Schnall et al. proposed a user-centred model for mHealth design based on iterative relevance, rigor, and design cycles [15]. These findings motivate the separation of functional verification from future end-user usability studies in the present work.

Usability evaluation is particularly relevant in health applications. Maramba et al. reviewed empirical usability methods in eHealth and identified questionnaires, task completion, think-aloud protocols, interviews, heuristic testing, and focus groups among the most frequently used approaches [16]. The System Usability Scale (SUS) has also been benchmarked specifically for digital health applications, providing a useful reference for future user studies [17]. In the current study, the objective is narrower: the reported experiments constitute technical and functional verification, while a caregiver-based usability study is reserved for future work.

Voice interaction is another relevant design dimension. Voice assistants are software agents that interpret speech and respond using synthesized speech, and their use has expanded across consumer devices [12]. Reviews of voice-assistant use among older adults indicate that usability and acceptance are common evaluation outcomes [13]. Voice interaction can therefore complement conventional graphical navigation, but its performance may be influenced by acoustic conditions, device configuration, and speech-recognition resources. For this reason, the present study evaluates a fixed command vocabulary experimentally rather than assuming that general speech recognition performance represents application-level command recognition.

3. Materials and Methods

3.1. Study Design

A functional experimental study was conducted to characterize the implemented Android application under disconnected conditions. The evaluation comprised five technical protocols: startup-time measurement, predefined voice-command recognition, offline functional verification, runtime resource profiling, and controlled battery-charge reduction measurement. The startup, functional, runtime, and battery protocols were conducted on the POCO M5s reference device, whereas the voice-command protocol was expanded to five Android devices and eight speakers to characterize variability across devices and speakers.

3.2. Digital Structuring of the Caregiving Content

The application was developed from the caregiving material associated with the project. According to the project documentation, the content comprises 127 caregiving processes organized into categories and steps, together with a glossary, images, audio elements, caregiver and patient information, a care log, and a calendar. The software project was implemented incrementally, with the mobile application developed in Unity and a local SQLite database containing the information required for standalone consultation.

The digital representation was designed around task-oriented navigation rather than open-ended clinical decision support. Each process is presented through a sequence of screens and can be reached through graphical controls or predefined voice commands. This approach is consistent with the need for clear information organization and accessible interaction in mHealth systems for older adults [8–11,15].

3.3. System Architecture

The mobile application follows a modular architecture comprising four principal components: (i) Unity-based user interface, (ii) application logic, (iii) local SQLite data management, and (iv) voice interaction. Separating presentation, application logic, data management, and interaction services supports maintainability and iterative software development [18,19]. The architecture is shown in Figure 1.

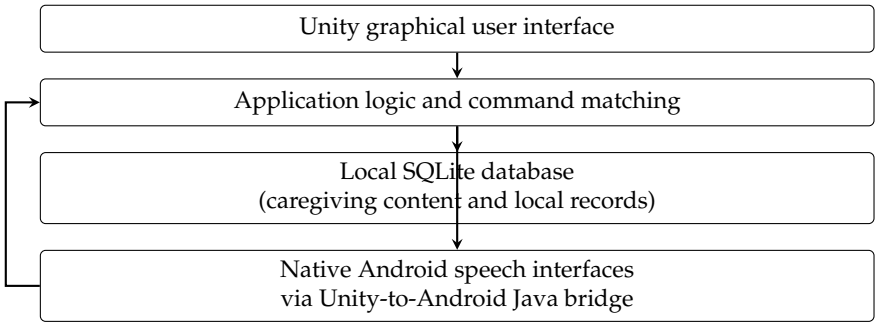


Figure 1. Compact architecture of the evaluated application. The local database stores the caregiving content and local records, while voice recognition and synthesis are accessed through native Android interfaces.

3.4. Database Design

A local SQLite database was implemented to store the caregiving content and application records. SQLite is an embedded, file-based database designed for applications where a separate database server is unnecessary [20]. The project documentation reports 22 tables, including tables associated with caregiving procedures, glossary information, records, audio, and voice commands. The present evaluation does not benchmark the database engine independently; its role is considered as the local persistence layer supporting offline operation.

3.5. Voice Interaction Module

Voice-to-text and text-to-speech were integrated through Android speech interfaces accessed from Unity using a Java bridge. The Unity C# components invoke the Android bridge to configure and control speech recognition and synthesis. After speech recognition, the returned text is evaluated against a predefined command vocabulary associated with application actions. Consequently, the system is not an open-domain conversational agent; it is a task-oriented voice-navigation mechanism. This design is consistent with the use of voice interaction as an additional interface rather than as a general clinical dialogue system [12–14,21–23].

The evaluated implementation uses the Android `SpeechRecognizer` interface through the Java bridge. The project source reviewed for this study does not explicitly set `RecognizerIntent.EXTRA_PREFER_OFFLINE`; therefore, offline operation is not attributed to that flag. Instead, disconnected operation was established experimentally by disabling network connectivity during the functional and voice protocols. Because Android speech recognition can depend on installed recognition services and language resources, the specific speech-service version and installed offline language package should be documented for each device in a future reproducibility protocol.

3.6. Offline Operational Model

The application stores the caregiving content, images, glossary information, and local records on the device. The project documentation explicitly describes SQLite as the local database used to avoid requiring Internet access for application content. The software project also includes server-side components for potential communication and monitoring, but these components are outside the scope of the present disconnected evaluation. Offline operation was therefore treated as an experimental property rather than inferred solely from the architecture.

3.7. Functional Evaluation Protocol

The application was installed on a POCO M5s smartphone running Android 13 with 8 GB of physical RAM. The device also had 3 GB of virtual memory expansion enabled; this value was not included as physical RAM. All tests were conducted with airplane mode enabled.

For startup time, 20 independent trials were recorded. The measured values were summarized using the arithmetic mean, minimum, maximum, and sample standard deviation. For voice recognition, ten predefined Spanish commands were evaluated by eight speakers on five Android devices, yielding 400 trials (8 speakers $\times$ 5 devices $\times$ 10 commands). The five devices were POCO M5s, POCO M8, POCO X3 Pro, Redmi Note 10, and Honor Magic 7. Each command was spoken once per device by each speaker. Two of the eight speakers were authors of the system and six were external participants. The participants were shown the command list before testing, and the commands were executed in the same order as recorded in the experimental dataset. The same application version was used on all devices. The devices were not connected to the Internet during the voice tests. Speech was captured using the device microphone; the speaker-to-device distance varied approximately from 10 to 50 cm, with approximately 20 cm as the target distance, and environmental background noise was present. A recognition was classified as correct when the spoken command was correctly identified and the application navigated to the corresponding procedure. Accuracy was calculated as the number of correct trials divided by the total number of trials.

A separate end-to-end latency protocol was subsequently analyzed from application logs. It comprised three successful measurements for each of the ten commands on each of the five devices, yielding a balanced dataset of 150 successful measurements. The latency interval was defined operationally from activation of speech recognition in the application to successful navigation to the corresponding procedure. The current integration did not instrument a speech-onset callback; therefore, this metric is reported as response latency from recognition activation rather than as physical speech-to-navigation latency. Five unsuccessful logged events were retained separately and were not substituted into the balanced successful dataset. Finally, thirteen application functions were tested in airplane mode and classified as PASS when the corresponding function could be accessed and completed as intended.

Runtime memory was characterized using Android's `dumpsys meminfo` command and total proportional set size (PSS). Twelve observations were recorded at the main menu and eight during navigation through application modules. CPU utilization was obtained using Android's `dumpsys cpuinfo` command, with ten observations at the main menu and ten during secondary-menu interaction. Battery-charge reduction was evaluated in three 30-min sessions under the same disconnected configuration; the initial and final battery percentages were recorded and their difference was used as the battery-charge reduction. End-to-end voice latency was analyzed from application log records using the separate 150-measurement dataset described above. SQLite query latency was measured independently using the database benchmark described below.

The evaluation follows the general principle that digital-health systems should be assessed using explicit, reproducible functional and usability measures [16,17,24]. The raw measurements used for the present analysis are provided as Supplementary File S1. Because the present experiment did not recruit caregivers as participants, it is reported as technical functional verification rather than a human-subject usability study.

3.8. Application Interface Evidence

Figure 2 presents representative screenshots of the implemented application taken from the project presentation. The screenshots illustrate the category menu and the result of a predefined voice command. They are included as evidence of the implemented interface rather than as decorative elements.

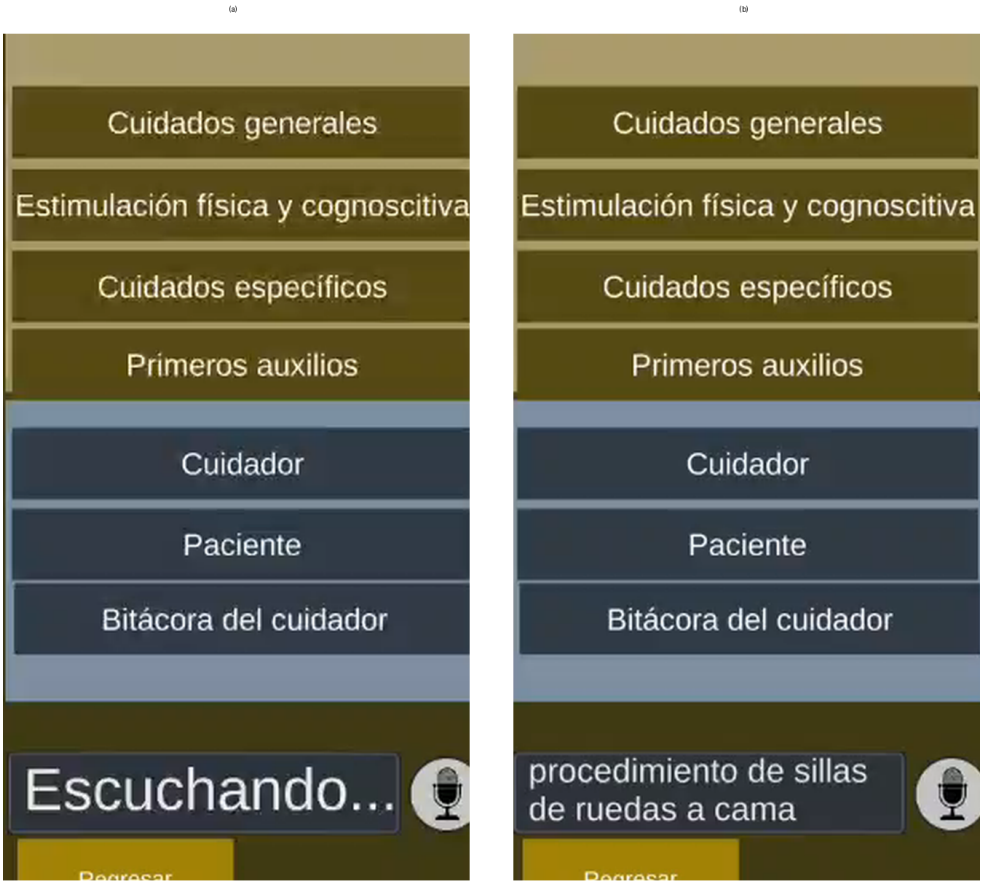


Figure 2. Representative screenshots of the implemented application: (a) category and caregiver/patient navigation menu; (b) predefined voice command recognized and displayed in the interface. The panels were cropped to remove device status bars and third-party/government branding that are not necessary for the technical analysis.

4. Results

4.1. Startup Time

Twenty startup trials were performed with the application in airplane mode. The measured values ranged from 3.9 to 4.1 s. The mean startup time was 4.00 s and the sample standard deviation was 0.03 s. The narrow range indicates stable startup behavior for the tested device and application build.

Table 1: Startup time measured over 20 trials.

Statistic	Result
Number of trials	20
Minimum	3.9 s
Maximum	4.1 s
Mean	4.00 s
Sample standard deviation	0.03 s

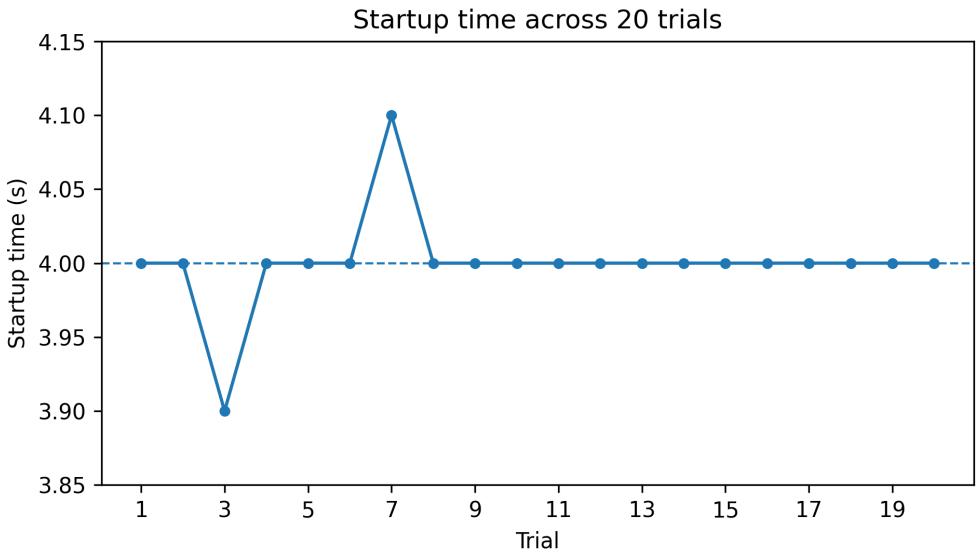


Figure 3. Startup time across the 20 individual trials. The dashed line indicates the sample mean of 4.00 s.

4.2. Voice Command Recognition

Across the five devices and eight speakers, 355 of 400 voice commands were correctly recognized, corresponding to an overall accuracy of 88.75%. The complementary error rate was 11.25%. A Wilson 95% confidence interval for the overall proportion was 85.28–91.49%. Because the same speakers and commands were evaluated across devices, these descriptive confidence limits should not be interpreted as accounting for repeated-measures clustering.

Table 2: Recognition accuracy for the ten predefined voice commands.

Command	Correct	Total	Accuracy
Higiene de manos	30	40	75.0%
Higiene de ojos	38	40	95.0%
Corte de uñas	33	40	82.5%
Pulso arterial	35	40	87.5%
Frecuencia respiratoria	34	40	85.0%
Colostomía	35	40	87.5%
Urostomía	34	40	85.0%
Cómodo	40	40	100.0%
De cama a silla	20	40	50.0%
Prótesis dental	36	40	90.0%
Total	355	400	88.75%

The highest command-level accuracy was obtained for “Cómodo” (100%), followed by “Higiene de ojos” (95%). The lowest value was observed for “De cama a silla” (50%), which accounted for 20 of the 45 total errors. The remaining commands ranged from 75.0% to 90.0% accuracy.

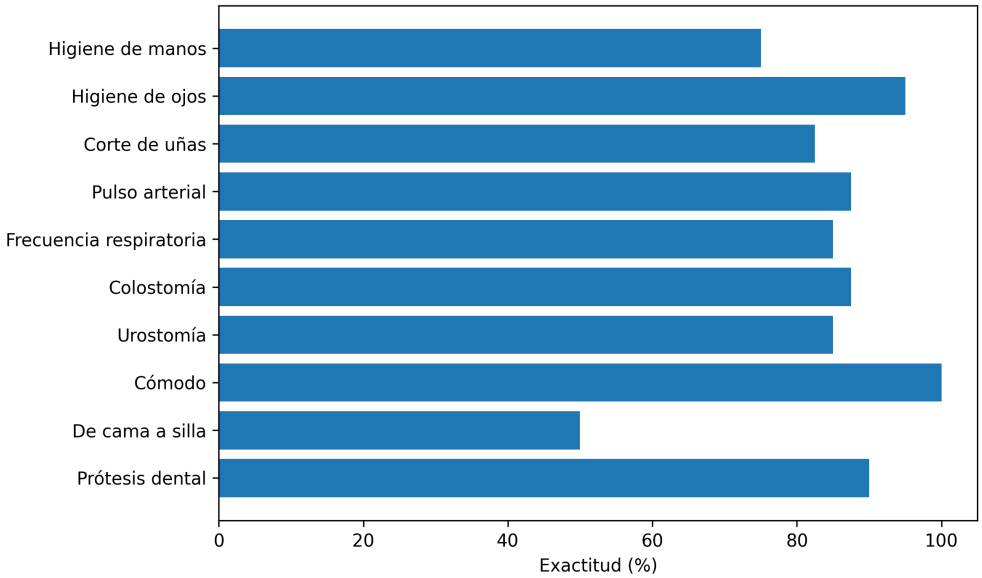


Figure 4. Recognition accuracy for the ten predefined voice commands across eight speakers and five Android devices.

4.2.1. Device-level results

The accuracy by device was 91.25% (73/80) for POCO M5s, 77.50% (62/80) for POCO M8, 80.00% (64/80) for POCO X3 Pro, 95.00% (76/80) for Redmi Note 10, and 100.00% (80/80) for Honor Magic 7. The mean device accuracy was 88.75% with a sample standard deviation of 7.32 percentage points.

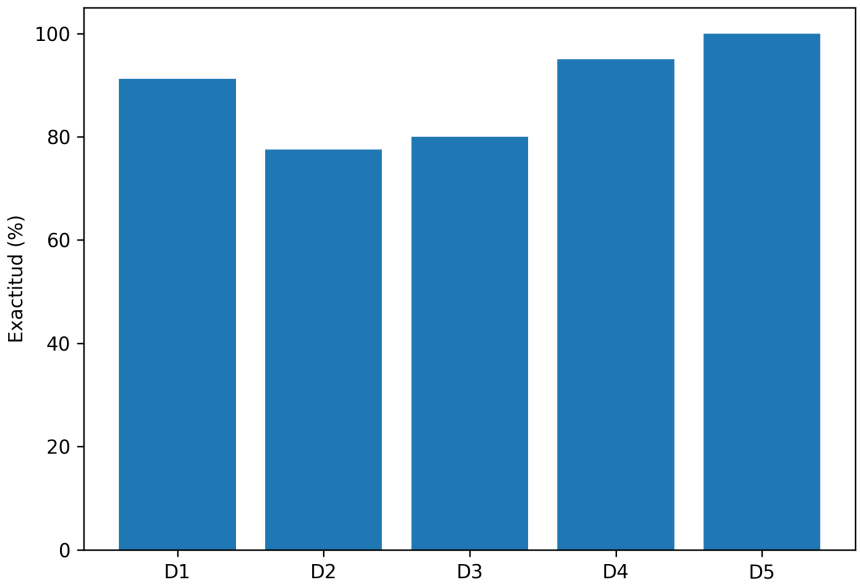


Figure 5. Voice-command accuracy by Android device.

4.2.2. Speaker-level results

Speaker accuracy ranged from 70% to 100%. Speakers 1 and 8 achieved 100%, whereas Speaker 2 had the lowest accuracy (70%). The two authors achieved 90/100 correct trials (90.00%), while the six external participants achieved 265/300 (88.33%). The difference between groups was 1.67 percentage points.

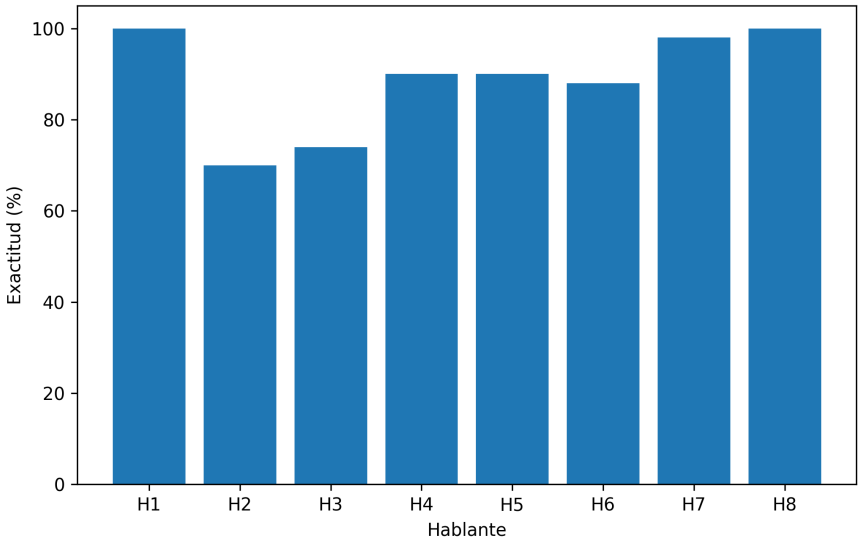


Figure 6. Voice-command accuracy by speaker. Speakers 4 and 5 were the authors of the system.

4.2.3. Exploratory analysis of command length

The number of words in the ten commands was examined as an exploratory descriptor. Pearson correlation between command length and accuracy was $r = -0.785$ ($p = 0.007$), while Spearman rank correlation was $\rho = -0.524$ ($p = 0.120$). The negative direction suggests that longer commands tended to have lower accuracy in this dataset; however, the analysis is based on only ten command-level observations and command length is confounded with lexical and phonetic characteristics. Therefore, the result is considered exploratory and does not establish that command length causes recognition errors.

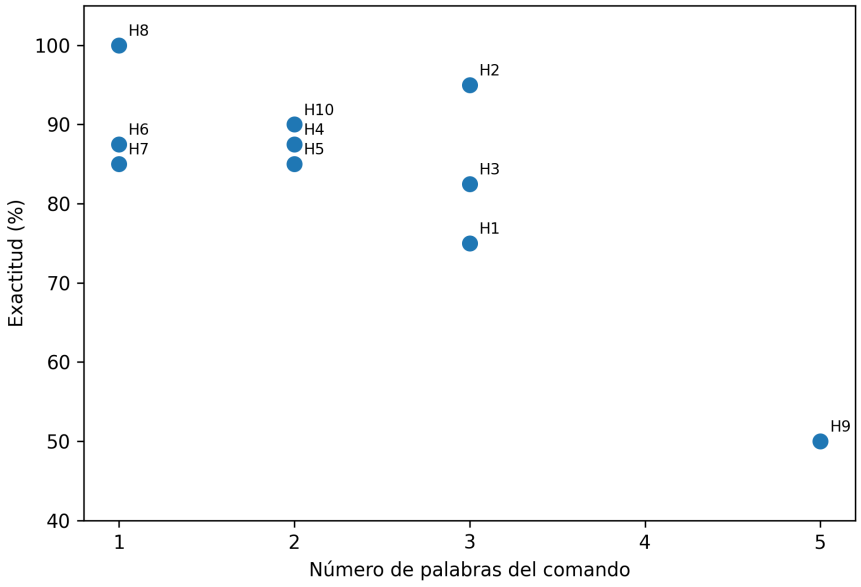


Figure 7. Exploratory relationship between number of words in each command and recognition accuracy. Labels identify the ten commands.

The voice protocol was performed without an active Internet connection. Thus, the 355 successful recognitions demonstrate that the evaluated command set remained operational in the disconnected configuration used for the experiment.

4.3. End-to-End Voice Response Latency

The balanced latency dataset contained 150 successful measurements (5 devices $\times$ 10 commands $\times$ 3 repetitions). Across all measurements, the mean latency was 3604.09 ± 768.84 ms, with a median of 3419.36 ms and a range of 2551.74–8979.48 ms. The device-level means ranged from 3288.17 ms to 3876.55 ms. The maximum value of 8979.48 ms was retained as an observed measurement rather than removed as an outlier.

Table 3: End-to-end voice response latency for the balanced successful dataset.

Device	n	Mean (ms)	SD (ms)	Median (ms)
Honor Magic 7	30	3288.17	431.04	3193.18
POCO M5s	30	3470.46	449.43	3406.41
POCO M8	30	3656.43	591.72	3633.10
Redmi Note 10	30	3728.83	898.88	3570.71
POCO X3 Pro	30	3876.55	1128.62	3538.33
All devices	150	3604.09	768.84	3419.36

The command-level means were 3114 ms for “Cómodo”, 3222 ms for “Corte de uñas”, 3263 ms for “Colostomía”, 3462 ms for “Higiene de ojos”, 3544 ms for “Urostomía”, 3580 ms for “Prótesis dental”, 3737 ms for “Higiene de manos”, 3976 ms for “Pulso arterial”, 4000 ms for “De cama a silla de ruedas”, and 4053 ms for “Frecuencia respiratoria”. These values are descriptive and should not be interpreted as causal effects of command length or lexical complexity.

Five unsuccessful logged events were excluded from the balanced successful-latency analysis and retained separately: “porte de uñas”, “hijita de manos”, “purotomía”, “prócesis dental”, and “por su alteriar”. Their presence illustrates that response latency and recognition success are distinct outcomes; unsuccessful recognition cannot be represented as a successful navigation latency without imposing an arbitrary endpoint.

4.4. Offline Functional Verification

Thirteen application functions were tested in airplane mode. The evaluated functions included startup, introduction, caregiving categories, procedures, images, glossary access, caregiver and patient information, care records, and calendar functions. All thirteen checks were completed successfully.

Table 4: Functional verification with the device in airplane mode.

Function	Result
Start	PASS
Introduction	PASS
General care	PASS
Stimulation	PASS
Specific care	PASS
First aid	PASS
Procedures	PASS
Images	PASS
Glossary	PASS
Caregiver	PASS
Patient	PASS
Care log	PASS
Calendar	PASS
Total	13/13 (100%)

4.5. Runtime Resource and Battery Evaluation

Runtime memory and CPU utilization were characterized on the same POCO M5s device used for the functional tests. At the main menu, 12 PSS observations produced

a mean of 355.10 ± 61.21 MB, with a range of 162.18–378.39 MB. During navigation through application modules, eight observations produced a mean of 384.31 ± 17.02 MB, with a range of 359.66–400.97 MB. The overall mean across the 20 PSS observations was 366.79 MB. The lower PSS observation at the main menu was retained because it was an actual measurement obtained under that experimental condition.

CPU utilization was 7.30% for all ten main-menu observations. During secondary-menu interaction, the ten observations yielded 8.46 ± 0.19%, with a range of 8.40–9.00%. These values characterize process CPU utilization at the sampling moments and should not be interpreted as whole-device processor utilization or electrical power consumption.

Battery-charge reduction was measured in three 30-min sessions. The recorded reductions were 5, 6, and 5 percentage points, respectively, resulting in a mean reduction of 5.33 ± 0.58 percentage points per 30-min session, equivalent to approximately 10.67 percentage points per hour under the tested conditions. This is a charge-percentage measurement rather than a direct electrical energy measurement.

Table 5 summarizes the experimentally measured indicators, including the separately analyzed end-to-end voice-response latency.

Table 5: Summary of experimentally obtained results.

Metric	Result
Startup time	4.00 ± 0.03 s
Voice recognition	88.75% (355/400)
End-to-end latency	3.604 ± 0.769 s (150 successful)
SQLite valid executions	1250/1250 (100%)
SQLite mean latency range	0.0985–1.8904 ms
Offline functional verification	13/13 (100%)
Mean PSS, all observations	366.79 MB
Battery reduction	5.33 ± 0.58 percentage points/30 min

5. Discussion

The results support three main observations. First, the application showed stable startup behavior on the tested POCO M5s, with a mean of 4.00 s and a small sample standard deviation across 20 trials. This measurement should be interpreted as a device-specific characterization rather than a benchmark applicable to all Android smartphones.

Second, the expanded voice-command evaluation achieved 88.75% accuracy across 400 trials involving eight speakers, five Android devices, and ten predefined commands. The results reveal variability at both the device and speaker levels. Device accuracy ranged from 77.50% to 100%, while speaker accuracy ranged from 70% to 100%. The two authors achieved 90.00% accuracy compared with 88.33% for the six external participants, a small descriptive difference of 1.67 percentage points. This comparison does not support a strong dependence on prior familiarity with the system, although the sample is too small for a definitive inferential conclusion.

A particularly relevant finding was the lower performance of the command “De cama a silla”, with 50% accuracy. The exploratory command-length analysis showed a negative association between the number of words and accuracy. However, because only ten commands were analyzed and lexical and phonetic complexity are confounded with length, the result should be treated as a hypothesis-generating observation rather than evidence of causality. The constrained vocabulary remains an important architectural feature because the application is intended to identify task-specific commands associated with procedure navigation rather than unrestricted conversational interaction.

Third, all thirteen functions included in the offline protocol were successfully executed with airplane mode enabled.

The end-to-end latency analysis provided an additional application-level measure that was not available in the earlier functional protocol. The mean of 3.604 s includes speech-recognition processing and the subsequent application navigation interval, but

it does not isolate physical speech onset, speech decoding time, or database access independently. The variability across devices, particularly the larger standard deviation observed for the POCO X3 Pro, indicates that response timing should be treated as an empirical characteristic of the tested configuration rather than as a device-independent property.

Fourth, the resource measurements indicate modest process CPU utilization and a higher memory footprint during module navigation than at the main menu. The mean PSS increased by approximately 29.21 MB between the two observed states. The battery test showed a mean reduction of 5.33 percentage points over 30 min under the controlled conditions. These measurements provide an initial device-level characterization, but they do not constitute a cross-device benchmark or a direct measurement of electrical power consumption. This is important because offline capability was evaluated experimentally rather than inferred only from the presence of SQLite. The result demonstrates that the content and functions included in the protocol were available without an active Internet connection on the tested device. The same disconnected configuration was used during the voice experiments, providing direct evidence that the evaluated command set also remained functional under that condition.

The findings are consistent with previous literature emphasizing structured content, accessible interfaces, and explicit usability evaluation for older-adult and caregiver-oriented mHealth applications [5–11,15–17]. However, the present study differs from usability or clinical-effectiveness studies because it does not claim improvement in caregiver outcomes, clinical outcomes, or user satisfaction. It establishes a technical foundation that can subsequently be evaluated with caregivers using standardized usability instruments such as SUS [17].

Several limitations should be considered. The startup, functional, runtime, and battery protocols used a single smartphone model and a single application build, while the voice experiment used five Android devices and eight speakers. The voice evaluation included one predefined command vocabulary and one utterance per command, so the results cannot be generalized to arbitrary speakers, speech patterns, or Android configurations. Microphone distance was controlled approximately but not instrumented continuously, and objective acoustic-noise level was not recorded. Offline speech recognition may depend on the recognition resources installed on the device; consequently, reproducibility should be assessed across devices, speech resources, and language configurations in future work. The battery experiment included only three 30-min sessions and used battery-percentage change rather than direct electrical power measurement. The latency protocol used application logs and therefore did not measure physical speech onset. The SQLite benchmark measured query execution rather than complete UI response time. No caregiver participants were included, and no clinical or usability outcome was measured.

Finally, the application should not be interpreted as a clinical decision-support system. Its purpose in the evaluated version is to provide structured caregiving information, procedure navigation, local records, and predefined voice interaction. Future validation with caregivers and health professionals is required before making claims about usability, safety, effectiveness, or impact on care outcomes.

6. Conclusion

This study presented the design and functional evaluation of an Android mobile application for caregivers of dependent older adults. The system was developed with Unity and a local SQLite database and organizes 127 caregiving processes together with supporting information such as procedures, images, glossary content, caregiver and patient records, care logs, and a calendar.

The experimental evaluation included a POCO M5s reference configuration running Android 13 with 8 GB of physical RAM, while the voice and SQLite protocols were expanded to five Android devices. Twenty startup trials produced a mean time

of 4.00 ± 0.03 s. The voice evaluation produced 355 correct recognitions out of 400 trials, corresponding to 88.75% overall accuracy across eight speakers, five Android devices, and ten predefined commands. The separate balanced latency dataset yielded a mean response latency of 3.604 ± 0.769 s across 150 successful measurements. Runtime profiling yielded 355.10 ± 61.21 MB PSS at the main menu and 384.31 ± 17.02 MB during navigation, while process CPU utilization was 7.30% and 8.46 ± 0.19%, respectively, for the tested states. Three 30-min battery sessions showed a mean reduction of 5.33 ± 0.58 percentage points. In addition, all thirteen application functions included in the offline verification protocol were successfully executed, corresponding to 100% functional success in the tested set.

These findings demonstrate that the evaluated application provided structured caregiving information and predefined voice navigation without an active Internet connection on the tested device. The evidence is limited to the tested hardware, Android configuration, command vocabulary, and functions included in the protocol. Nevertheless, the results support local data storage and constrained voice interaction as a practical architecture for caregiving tools intended for environments where reliable connectivity cannot be assumed.

7. Future Work

Although the proposed system demonstrates stable performance and full offline operability, several research and development extensions are planned to enhance its functionality and scientific contribution.

A primary direction involves conducting large-scale usability studies with caregivers across different regions of Mexico. Future evaluations will incorporate standardized usability assessment instruments, such as the System Usability Scale (SUS), to quantitatively measure user satisfaction, cognitive load, and long-term adoption patterns.

Another significant extension includes the integration of a statistical analysis module capable of extracting anonymized usage data for research purposes. This feature would allow identification of the most frequently consulted caregiving processes, interaction patterns, and workflow bottlenecks, contributing to evidence-based refinement of caregiving protocols.

From a technological perspective, future versions will explore the incorporation of artificial intelligence techniques to provide adaptive recommendations based on patient profiles and historical caregiving interactions. Lightweight machine learning models operating locally on-device will be prioritized to preserve offline operability and data privacy.

Cross-platform deployment is also planned, particularly expansion to iOS environments to increase accessibility. This will require architectural adaptation while preserving modular design principles.

Additionally, future work will focus on strengthening data security mechanisms through encrypted local storage and secure authentication layers, ensuring compliance with healthcare data protection standards.

Finally, integration with public health systems and potential interoperability with electronic health record (EHR) infrastructures in Mexico represents a long-term research objective. Such integration would enable coordinated care models while maintaining patient data sovereignty.

These future developments aim to transform the current system from a structured digital caregiving tool into an intelligent, scalable, and interoperable mHealth platform aligned with the evolving needs of the Mexican healthcare ecosystem.

Supplementary Materials: The following supplementary material is available online at the journal website: Supplementary File S1: Raw experimental measurements for startup, voice-command recognition, and airplane-mode functional verification.

Author Contributions: Conceptualization, I.S, C.P.O., and J.D.M.S; methodology I.S., C.P.O, and F.M.; formal analysis, I.S., C.P.O., F.M.. and J.D.M.S.; investigation, G.A.R., F.M., and C.P.O.; writing—original draft preparation, I.S., C.P.O., and G.A.R.; writing—review and editing, I.S., C.P.O., and J.D.M.S.; supervision, F.M., and C.P.O. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement: Not applicable. The study evaluated the software and its operation on a smartphone and did not involve human participants.

Informed Consent Statement: Not applicable.

Funding: Not applicable

Data Availability Statement: The raw experimental measurements used in this study are provided as Supplementary File S1. The application build and source materials are available from the corresponding author upon reasonable request.

Acknowledgments: We are thankful for the Artificial Vision Laboratory CIO in Aguascalientes.

Conflicts of Interest: The authors state that they have no competing financial interests or known personal relationships that could have influenced the work reported in this paper.

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